Chapter 1. The Pigeon-Hole Principle

Seven Is More Than Six

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1.1 The basic pigeon-hole principle

Motivation. "Seven is more than six" sounds empty, yet comparing the number of objects with the number of places they can go yields strong existence statements with almost no computation. The art is to decide what the balls and what the boxes are.

Theorem 1.1 (Pigeon-hole principle)

Let $n > k$ be positive integers. If n identical balls are placed into k identical boxes, some box receives at least two balls.

Proof. Suppose no box has two balls. Then every box has 0 or 1 ball; if $m \geq 0$ boxes are empty, the total number of balls is $k - m \leq k < n$, a contradiction. ■

Example 1.2

Some element of $7, 77, 777, \ldots$ is divisible by 2023.
Solution. Suppose none of the first 2023 elements $a_1, \ldots, a_{2023}$ is divisible by 2023. Their remainders lie in $\{1, \ldots, 2022\}$, so by Theorem 1.1 two of them, a_i and a_j with $i < j$, have the same remainder, and $2023 \mid a_j - a_i$. Now $a_j - a_i$ consists of $j - i$ sevens followed by i zeros, i.e. $a_j - a_i = a_{j-i} \cdot 10^i$. Since $\gcd(10^i, 2023) = 1$, $2023 \mid a_{j-i}$, contradicting the assumption. (Remainders are the boxes, the elements the balls.)

Example 1.3

In a round-robin chess tournament with n players, at any moment two players have finished the same number of games.
Solution. The possible numbers of finished games are $0, 1, \ldots, n-1$: n boxes for n players. But 0 and $n-1$ cannot occur simultaneously: a player who has finished all $n-1$ games has played everybody. So at most $n-1$ values occur, and Theorem 1.1 applies.

1.2 The generalized pigeon-hole principle

Motivation. If there are many more balls than boxes, some box must contain many balls. This quantitative version is just as easy to prove and is the form used in geometry, number theory and graph theory.

Theorem 1.4 (Pigeon-hole principle, general version)

Let n, m, r be positive integers with $n > rm$. If n identical balls are distributed into m identical boxes, some box receives at least $r + 1$ balls.

Proof. Otherwise each box holds at most r balls, so all boxes together hold at most $rm < n$ balls, a contradiction. ■

Example 1.5

Ten points lie in a unit square. Then two of them are at distance < 0.48, and three of them can be covered by a disk of radius 0.5.

Solution. Cut the square into nine squares of side $\frac{1}{3}$ (left figure). Some small square contains two points (Theorem 1.1), and their distance is at most the diagonal $\frac{\sqrt{2}}{3} < 0.48$. Cut the square by its diagonals into four triangles (right figure). Since $10 > 2 \cdot 4$, some triangle contains three points (Theorem 1.4). The circumradius of that right isosceles triangle is half its hypotenuse, $\frac{1}{2}$, so a disk of radius 0.5 covers it.

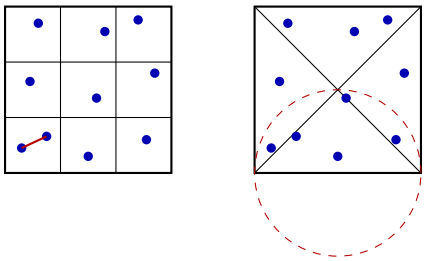


Figure. *Example 1.5: nine small squares force two close points; four triangles force three points under one disk of radius $\frac{1}{2}$ (dashed: the disk around the bottom triangle).*

Example 1.6

Within the last 1000 years, the reader had an ancestor A such that some person P was an ancestor of both parents of A.

Solution. With 25 years per generation, 1000 years are 40 generations, and the family tree of the reader has $1 + 2 + \cdots + 2^{40} = 2^{41} - 1$ nodes. If two nodes are the same person B, then B serves as P (for the nearest common descendant). Otherwise there would be $2^{41} - 1$ distinct people, but fewer than $40 \cdot 10^{10} < 2^{41} - 1$ people lived during those 40 generations (the population never exceeded 10^{10}), a contradiction.

Example 1.7 (The handshake party)

Mr. and Mrs. Smith invite four couples. Acquaintances shake hands (nobody with themself or their spouse). Afterwards Mr. Smith observes that, apart from himself, nobody shook hands the same number of times. How many hands did Mrs. Smith shake?

Solution. Model people as vertices of a **graph**, joining two vertices by an edge if those people shook hands. The nine people other than Mr. Smith have nine distinct handshake numbers in $\{0, \ldots, 8\}$, so every value occurs once; let Y_i be the person with i handshakes.

Y_8 shook hands with everyone except one person, who must be the spouse; since Y_0 shook no hands, Y_8 and Y_0 are married.

Y_7 missed exactly two people, one of them the spouse. These must be Y_0 (no handshakes) and Y_1 (whose only handshake was with Y_8), so the spouse of Y_7 is Y_1.

Continuing in the same way, Y_6–Y_2 and Y_5–Y_3 are couples, so Y_4 is Mrs. Smith: she shook hands **four** times. By Example 1.3, a party in which everyone shakes a different number of hands is impossible; exempting one person leaves exactly one possibility, shown below.

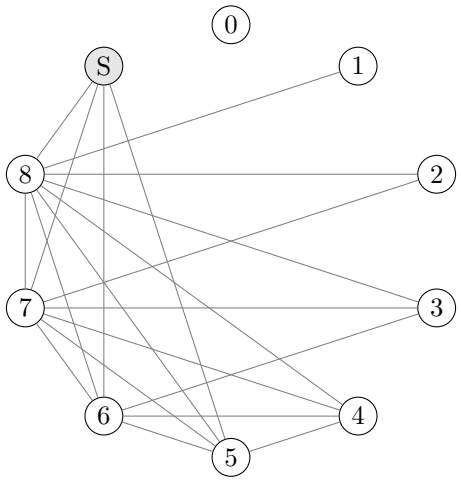


Figure. *The only possible party of Example 1.7: vertex i shook i hands, S is Mr. Smith. Vertex $k \geq 5$ is joined to S and to all $j \geq 9-k$; the couples are $\{8,0\},\{7,1\},\{6,2\},\{5,3\}$ and $\{4,S\}$.*

The notion of a graph will be central in Chapters 9–12.

1.3 Selected exercises

Motivation. The exercises below, taken from the book, train the one real skill of this chapter: choosing the boxes so cleverly that a bare count of balls and boxes gives the conclusion.

Exercise 1.11 (Sum or difference divisible by 1000)

Prove that among any 502 positive integers there are two whose sum or difference is divisible by 1000.

Solution.

Step 1 (boxes). For $r \in \{0,1,\dots,999\}$ let $\bar{r}$ denote the residue class of r modulo 1000. Form the 501 boxes

$$\{\bar{0}\},\quad \{\bar{500}\},\quad \{\bar{1},\overline{999}\},\quad \{\bar{2},\overline{998}\},\quad \dots,\quad \{\overline{499},\overline{501}\},$$

that is, $\{\bar{r},\overline{1000-r}\}$ for $1 \leq r \leq 499$ together with the two singletons. Every residue class lies in exactly one box.

Step 2 (pigeon-hole). Put each of the 502 integers into the box of its residue class. Since $502 > 501$, Theorem 1.1 gives two distinct integers a, b (distinct as members of the list) in the same box.

Step 3 (conclusion). If $a \equiv b \pmod{1000}$, then $1000 \mid a-b$. Otherwise the box is $\{\bar{r},\overline{1000-r}\}$ with $a \equiv r$ and $b \equiv 1000-r$ (or the reverse), so $a+b \equiv 1000 \equiv 0$ and $1000 \mid a+b$. (In the singleton boxes only the first case occurs.) ∎

Exercise 1.15 (Six goals in consecutive games)

A soccer team played 30 games and scored 53 goals, scoring at least one goal in each game.
(a) Prove that there was a sequence of consecutive games in which the team scored exactly six goals.
(b) Prove that the claim of part (a) can fail if the team scored 60 goals, the other conditions unchanged.

Solution.

(a) Let s_k be the number of goals scored in games $1, \dots, k$. Since every game has at least one goal,

$$1 \le s_1 < s_2 < \cdots < s_{30} = 53.$$

Consider the 60 numbers $s_1, \dots, s_{30}, s_1 + 6, \dots, s_{30} + 6$. All of them lie in $\{1, 2, \dots, 59\}$, so by Theorem 1.1 two of them are equal. The s_k are pairwise distinct, and so are the $s_k + 6$; hence $s_j = s_i + 6$ for some i, j. Then $s_j > s_i$, so $j > i$, and games $i+1, \dots, j$ produced exactly $s_j - s_i = 6$ goals.

(b) Let the goals per game be $1, 1, 1, 1, 1, 7$, repeated five times: 30 games, $5 \cdot 12 = 60$ goals, at least one goal per game. A block of consecutive games that contains a game with 7 goals has at least 7 goals. A block that contains no such game lies within one run of five 1's and has at most 5 goals. So no block has exactly 6 goals. ■

Exercise 1.18 (Two 1's in north-east position)

Let $n \ge 2$ and let A be an $n \times n$ matrix with entries 0 and 1 only, at least $2n$ of which are 1. Prove that A contains two entries equal to 1 such that one of them is strictly above and strictly to the right of the other.

Solution. Index the entries by (i, j), row i counted from the top and column j from the left. The entry (i', j') is strictly above and strictly to the right of (i, j) iff $i' < i$ and $j' > j$.

Step 1 (boxes). For $s = 2, 3, \dots, 2n$ let $D_s = \{(i, j) : i + j = s\}$ be the s-th anti-diagonal. These $2n - 1$ sets partition the positions of A.

Step 2 (pigeon-hole). There are at least $2n > 2n - 1$ entries equal to 1, so by Theorem 1.1 two of them, $(i, j) \ne (i', j')$, lie in the same D_s.

Step 3 (conclusion). From $i + j = i' + j'$ and $(i, j) \ne (i', j')$ we get $i \ne i'$; say $i' < i$. Then $j' = s - i' > s - i = j$, so (i', j') is strictly above and strictly to the right of (i, j).

The bound is sharp: the $2n - 1$ ones of the last row and the first column contain no such pair: two of them share a row or a column, or they are $(i, 1)$ and (n, j) with $i < n$ and $j > 1$, and then the upper one is to the left. ■

Exercise 1.35 (One number divides another)

(Part (b) of the book's exercise.) Let $n \ge 1$. Prove that among any $n + 1$ distinct integers selected from $[2n]$ there are two such that one is a multiple of the other.

Solution.

Step 1 (boxes). Every positive integer x can be written uniquely as $x = 2^a m$ with $a \ge 0$ and m odd (divide by 2 as long as possible). Call m the odd part of x. For $x \in [2n]$ the odd part lies in $\{1, 3, \dots, 2n - 1\}$, a set of n elements.

Step 2 (pigeon-hole). The $n + 1$ selected integers have their odd parts among n values, so by Theorem 1.1 two different selected integers $x = 2^a m$ and $y = 2^b m$ have the same odd part m.

Step 3 (conclusion). Since $x \ne y$, $a \ne b$; say $a < b$. Then $y = 2^{b-a} x$ is a multiple of x.

The number $n + 1$ cannot be lowered: the n integers $n + 1, \dots, 2n$ contain no such pair, because twice the smallest of them already exceeds $2n$. ■

Exercise 1.37 (Multiples of an irrational number)

Let r be an irrational real number. Prove that there is a positive integer n such that the distance of nr from the nearest integer is less than 10^{-10}.

Solution. Write $N = 10^{10}$ and $\{x\} = x - \lfloor x \rfloor \in [0,1)$ for the fractional part.

Step 1 (boxes). Split $[0,1)$ into the N half-open intervals $I_a = \left[\frac{a}{N}, \frac{a+1}{N}\right)$, $a = 0, \dots, N-1$.

Step 2 (pigeon-hole). The $N+1$ numbers $\{kr\}$, $k = 0, 1, \dots, N$, lie in $[0,1)$, so by Theorem 1.1 two of them, $\{kr\}$ and $\{\ell r\}$ with $0 \le \ell < k \le N$, lie in the same I_a, hence $\left|\{kr\} - \{\ell r\}\right| < \frac{1}{N}$.

Step 3 (conclusion). Put $n = k - \ell \ge 1$ and $M = \lfloor kr \rfloor - \lfloor \ell r \rfloor \in \mathbb{Z}$. Then

$$|nr - M| = \left|(kr - \lfloor kr \rfloor) - (\ell r - \lfloor \ell r \rfloor)\right| = \left|\{kr\} - \{\ell r\}\right| < 10^{-10},$$

so the distance of nr from the nearest integer is less than 10^{-10}. Irrationality guarantees in addition that this distance is positive, so the multiples of r come arbitrarily close to integers without hitting them. ∎